Purification of P-glycoprotein from KB-A1 HeLa Cells Using Preparative LDS Gel Electrophoresis

Contributed by Carrie A. Paquette and Stan Ivey, Department of Biology, Delaware State University, Dover, DE 19901.

Introduction

P-glycoprotein is a membrane glycoprotein responsible for the intracellular removal of a broad range of cytotoxic agents including those used in the chemotherapeutic treatment of cancer. P-glycoprotein acts as an energy-dependent efflux pump that removes cytotoxins from multi drug resistant neoplastic cells (Weinstein *et al.*, 1991). Antibodies have been used extensively to identify and to help characterize this important membrane glycoprotein (Grogan *et al.*, 1990). Most of the antibodies raised against P-glycoprotein have been made to peptide sequences within the protein. JSB-1, a mAb raised to a carboxy terminus peptide sequence in the 170 kD P-glycoprotein isoform, was used in this study to identify P-glycoprotein on Western blots. To increase immunostaining sensitivity on our blots and to cut cost, we elected to raise polyclonal antibodies to purified P-glycoprotein, isolated using Bio-Rad's Model 491 Prep Cell.

P-glycoprotein is overexpressed in KB-A1 HeLa cells, a human cell line 100 times more resistant to adriamycin (a common anticancer drug used in chemotherapy) than normal HeLa cells. These cells offered an excellent source for human P-glycoprotein. P-glycoprotein was isolated from solubilized KB-A1 membranes and used to produce rabbit polyclonal antibodies to human P-glycoprotein. These polyclonal antibodies have greatly enhanced our detection of P-glycoprotein on blots. We report here the rapid isolation of human P-glycoprotein for biochemical characterization and antibody production.

Methods

Sample Preparation

The KB-A1 HeLa cell line was a gift from Dr. T. R. Tritton, University of Vermont. Cells were grown to 95% confluency in the presence of adriamycin and collected by trypsinization. Cells (2.2×10^6 cells) were washed in PBS and resuspended in lysing buffer (5.0 mM EDTA and 5.0 mM HEPES, pH 7.5, containing protease inhibitors) and placed on ice with slow stirring for 15 min. Lysed cells were centrifuged (45,000 x g) for 30 min at 4 °C. The pelleted cell membranes were solubilized in 0.5 ml of 3.5% w/v LDS (lithium dodecyl sulfate), 5.0 mM β-mercaptoethanol, and 60 mM Tris-HCl, pH 6.8. The solution was again centrifuged at 45,000 x g to remove any unsolubilized material. The supernatant fluid was heated for 3 min at 80 °C and cooled to room temperature. Glycerol was added to a final concentration of 10%. A protein profile of the solubilized cell lysate can be seen in Figure 1.

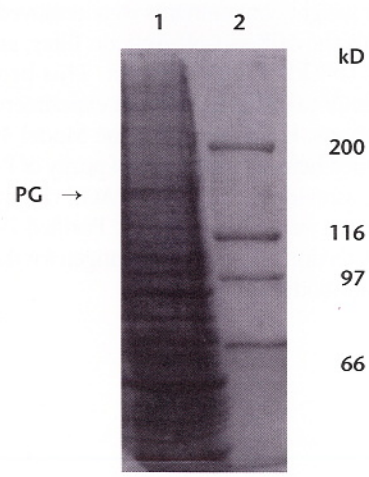

Fig. 1. Coomassie-stained protein profile of KB-A1 cell lysate separated on a 6.0% LDS-PAGE Mini-PROTEAN® II gel. Lane 1 contains solubilized membrane proteins isolated as described in Method. Lane 2 contains standard protein markers. The location of P-glycoprotein is indicated.

Preparative LDS -PAGE

All gel runs were done at 4 °C in a chromatography refrigerator. LDS-preparative gels and vertical analytical LDS gels were run using the buffer system described by Laemmli (1970), with LDS substituted for SDS to prevent the precipitation of detergent during the 4 °C running conditions. We found that running the preparative gel at 4° C produced cleaner separation of protein bands and prevented excess degradation during sample collection than room temperature runs. Sample (0.5 ml) was layered onto a 3.5% polyacrylamide stacking gel (1.0 cm) with a 7.0% polyacrylamide separating gel (2.6 cm) poured in the 37 mm diameter Model 491 Prep Cell gel tube assembly.

The upper, lower, and elution chambers were filled with chilled Tris/Glycine/LDS running buffer, pH 8.3. To maintain the 4 °C environment around the gel, the lower chamber buffer was continuously circulated using the accompanying Bio-Rad circulation pump. The sample was electrophoresed through the gel at 30 mA constant current for 15 hours.

Sample Collection and Analysis

Fractions were collected after the bromophenol blue marker dye eluted from the gel (about 1.0 h run time). An ISCO UA-5 Absorbance Detector with a flow rate of 1.0 ml/min was used to detect protein elution; 108 tubes were collected, each containing 8.0 ml of eluant. Every 5 tubes were pooled and concentrated to 0.5 ml using an Amicon Cell with a 50,000 MW cutoff cellulose acetate filter. Aliquots (40 µl) were removed from each concentrate for detection of P-glycoprotein using analytical LDS-PAGE (Figure 2). To locate the P-glycoprotein in the fractions, gels were stained with Coomassie® brilliant blue. P-glycoprotein was detected on accompanying Western blots immunostained using the monoclonal antibody, JSB-1 (purchased from Signet Laboratories, Inc.).

Results

Fractions 40–44, containing purified P-glycoprotein, eluted between 5 and 6 hours run time. Most breakdown products or low molecular weight contaminants were removed during concentration with the 50K cut-off Amicon filter, enhancing the purity of the 163 kD P-glycoprotein. This protein proved extremely difficult to isolate in our past experiments using conventional methods. However, using the Model 491 Prep Cell method described here, a high degree of purity of P-glycoprotein was obtained, as evidenced by LDS-PAGE (Figure 2, lane 3). Protein recoveries were not calculated. Purified P-glycoprotein was used for detection on gels and as antigen for the production of polyclonal antibodies.

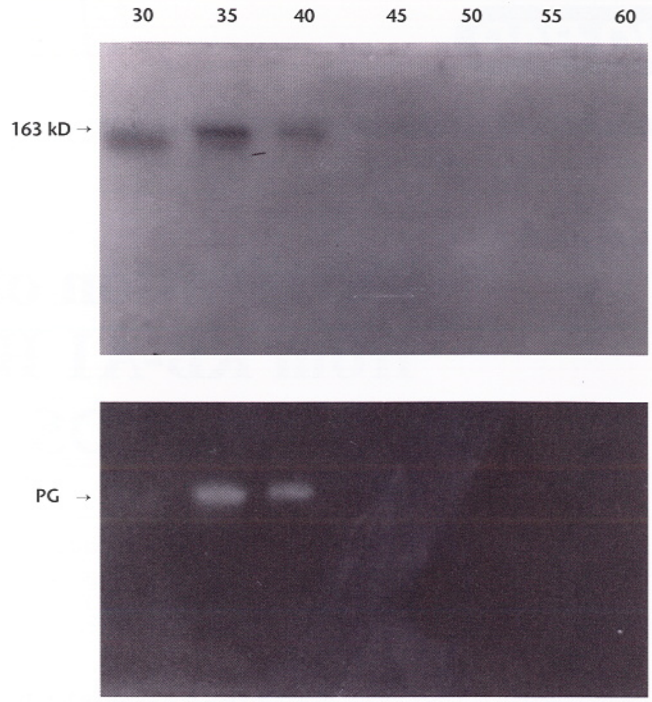

Fig. 2. Aliquots from Model 491 Prep Cell fractions 30–64 identifying isolated human P-glycoprotein. The top panel is a Coomassie-stained 6.0% LDS-PAGE mini gel. Each lane contained 40 µl of sample taken from pooled (5 tubes, each containing 8.0 ml) and concentrated (0.5 ml final volume) fractions as described in Methods. Lane 2 contained P-glycoprotein and a contaminant. Lane 3 contained the purified P-glycoprotein, having an apparent molecular weight of 163 kd. The bottom panel is a sister Western blot immunostained with JSB-1 mAb (Signet Labs, Inc., Dedham, MA) and a Bio-Rad second GAM-AP antibody Immun-Lite™ chemiluminescent assay kit. P-glycoprotein was detected in fractions 35–44 (lanes 2 and 3).

References

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